

Clinical Letter

Natatorial mishap: Skin lesions caused by jellyfish stings

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Dear Editors,

We report on two patients who developed painful skin lesions following jellyfish stings.

After swimming in the Bay of Lübeck three hours earlier, a 54-year-old man presented to our outpatient clinic with ill-defined, erythematous, mildly urticarial painful skin lesions. He reported that he had come into contact with a lion's mane jellyfish (giant jellyfish) while swimming. We treated him with topical corticosteroids and an oral antihistamine. All skin lesions healed completely within 24 hours.

Presenting to our outpatient clinic, a 55-year-old female patient reported to have come into contact with a jellyfish in Phuket (Thailand) eight days earlier. She could not name the exact species, though. Immediately after exposure, she had experienced burning pain. While rubbing with sand had initially led to symptom relief, the burning pain had recurred with a vengeance five days later. At that point, the patient had noted around 200 tiny "barbs" in her skin and had subsequently removed them with tweezers. Clinical examination revealed serpiginous, papular/urticarial skin lesions on the right side of her abdomen and on her right hand (Figure 1). On dermoscopy, we saw dark foreign bodies in the center of the papules (Figure 2). Topical treatment with a corticosteroid cream led to rapid clinical improvement. Eight months after the incident, the patient still showed mild hyperpigmentation.



Figure 1 Serpiginous erythematous, papular/urticarial skin lesions on the abdomen eight days after a jellyfish sting.



Figure 2 Dermoscopy: Residual nematocysts in the skin of the right hand eight days after a jellyfish sting.

Jellyfish are cnidarians and can be divided into three classes: Cubozoa (box jellyfish), Hydrozoa (hydrozoans) and Scyphozoa (true jellyfish) [1] (Table 1). Cubozoa include the sea wasp and the various Irukandji jellyfish. The most important representative of the Hydrozoa class is the Portuguese man o' war. Scyphozoa comprise the lion's mane jellyfish and the compass jellyfish. The jellyfish most commonly found along Germany's coast (North and Baltic Seas) are of the Scyphozoa class; they are only mildly venomous. In contrast, the various species of Cubozoa and the Portuguese man o' war, which can be very dangerous to humans, are primarily found in tropical waters [1].

While our first patient came into contact with a lion's mane jellyfish, we were unable to conclusively determine the species in the second case. However, we suspected a true jellyfish (Scyphozoa) to have been involved in this case "as well".

Jellyfish have an umbrella-shaped body and tentacles. The tentacles harbor specialized stinging cells (cnidocytes). A hair-like sensory process (cnidocil) projects from the surface of these cells. Each cnidocyte contains a bulb-shaped capsule (nematocyst) that consists of a coiled hollow barbed tubule. Following stimulation of the cnidocil, the coiled tubule is forcefully ejected from the capsule, and the venom contained therein is injected into the prey [2, 3]. Depending on the type of venom, clinical manifestations range from local reactions to systemic, life-threatening symptoms [2]. Jellyfish venoms that contain pore-forming proteins are particularly harmful, as they directly lead to cell death by disturbing the osmotic equilibrium; arrhythmia due to electrolyte imbalance may also occur [4]. Acute skin reactions can be attributed to direct toxic effects of the jellyfish venom [5]. Moreover, there have been reports of patients experiencing IgE-mediated anaphylaxis and type IV hypersensitivity reactions following

Table 1 Classification of jellyfish and associated symptoms (modified after [1, 2, 4]).

Class and species of jellyfish	Occurrence	Special features	Symptoms	Recommended treatment
Cubozoa (box jellyfish)				
Sea wasp (<i>Chironex fleckeri</i>)	Northern Australia	– Up to 30 cm in diameter – Tentacles 2–3 m long – Prefers shallow water	– Initially, severe pain to touch – Severe local reaction with “frosted cross-hatched marks” caused by intermittent necrosis – Cardiovascular depression with cardiac arrest	– Supportive measures – Acetic acid 4–6 % – Cardiopulmonary resuscitation – Antivenom
Irukandji (<i>Carukia barnesi</i>)	Northern Australia	– 1–2 cm in diameter – Tentacles 35 cm long	– Initial contact often goes unnoticed; frequently, delayed symptoms – Severe local reaction with “frosted cross-hatched marks” caused by intermittent necrosis – “Irukandji syndrome” with massive catecholamine release, severe pain, hypertension, pulmonary edema	– Supportive measures – Acetic acid 4–6 % – Cardiopulmonary resuscitation
Hydrozoa (hydrozoans)				
Portuguese man o’ war (<i>Physalia physalis</i>)	Pacific, Atlantic, Caribbean	– Shape resembles a ship at full sail – Single tentacle – Thermolabile venom	– Initially, severe pain to touch – Few, streaky lesions due to single tentacle – Severe local reactions – Cardiovascular reactions – Anaphylaxis	– Supportive measures – Heat application – Aluminum sulfate
Scyphozoa (true jellyfish)				
Lion’s mane jellyfish (<i>Cyanea capillata</i>)	Atlantic, North Sea, Baltic Sea	– Up to 2 m in diameter – Tentacles up to 30 m long – Given their intense color, tentacles are reminiscent of a “lion’s mane”	– Moderate pain to touch – Severe local reactions	– Supportive measures – Rinsing with seawater – Cold application – Neutralization with baking soda or shaving foam (topical agents containing potassium)
Compass jellyfish (<i>Chrysaora hysoscella</i>)	Atlantic, North Sea, Baltic Sea, Mediterranean	– 25 cm in diameter – Distinct, radially arranged brown markings	– Moderate pain to touch	– Supportive measures – Rinsing with seawater – Cold application

repeated jellyfish stings [5]. Chronic granulomatous dermatitis has also been reported to occur [6]. The tubules contained in the nematocysts of dangerous jellyfish are longer than those of less harmful species. The deeper injuries caused by the former might thus explain why stings caused by venomous jellyfish are also more painful [7]. Remarkably, functioning nematocysts can be found even in beached jellyfish [3]. The biphasic disease course in our second patient may possibly be explained by a delayed-type reaction to jellyfish remnants left in the epidermis. It is noteworthy that contact with oak processionary moth caterpillars may lead to clinically and histologically similar lesions as jellyfish stings [8].

Current treatment recommendations for acute jellyfish stings are primarily based on case reports and small case series. Provided the jellyfish sting occurred in the sea, the injured person should be brought to land immediately. If systemic symptoms occur, stabilization of vital functions is paramount. Any jellyfish remnants should be carefully removed from the skin [4]. Given that only 10–20 % of nematocysts are actually activated during contact with a jellyfish [3], measures should be taken to inactivate any remaining nematocysts; these measures vary depending on the type of jellyfish involved (Table 1). Treatment with freshwater should be avoided, as residual nematocysts may be activated by the hypotonic milieu [4]. It has been shown that the nematocysts of box jellyfish may be successfully inactivated by dowsing the area affected with acetic acid (4–6 %) for 30 seconds [4]. The Australian sea wasp produces a pore-forming venom that may lead to arrhythmia. Some have therefore proposed the use of calcium channel blockers in this context. However, given their questionable effectiveness and the risk of hypotension, these agents are no longer recommended today [5]. There is an antivenom available for sea wasp stings. Harvested from the serum of hyperimmunized sheep, it is administered intravenously in the event of life-threatening cardiovascular reactions [2, 4]. Compression bandages are contraindicated as any compression may lead to physical activation of remaining nematocysts [4]. Injuries caused by the Portuguese man o' war should be treated with warm water (42–45°C, for 20 minutes), as the venom is known to be thermolabile [2]. Application of aluminum sulfate 20 % has also been shown to inactivate any residual nematocysts of this species [2, 4]. By contrast, cold application is recommended for all other types of jellyfish [4]. Topical tacrolimus is a therapeutic option for patients with protracted granulomatous skin reactions [6]. In regions where the risk of severe jellyfish injuries is high, protective measures should be implemented

to avoid direct contact. Protective Lycra® suits have proven to be particularly useful in this regard [3].

Conflicts of interest

None.

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